

Inverse design of nonlinear mechanical metamaterials via video denoising diffusion models

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Supplementary information for ‘Inverse-design of nonlinear mechanical metamaterials via video denoising diffusion models’

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S0 Overview

We here provide further details regarding the data generation procedure, explaining the methods employed for creating the metamaterials under consideration, and the finite element method (FEM) setup to evaluate the nonlinear mechanical response of unit cells. We further present an extended formal background of the denoising diffusion framework and the corresponding deep neural network model architecture as well as the training and sampling protocol. We further describe in detail the evaluation of the generated structures with regard to the predicted full-field responses and corresponding stress-strain curves. Note that certain figures have been reproduced from the Extended Data of the main article—here with additional explanations—for ease of reading and completeness.

S1 Data generation process

S1.1 Design generation

We provide the pseudocode for the design generation process as described in the Methods section of the main article in Algorithm 1. We abbreviate the discrete Fourier transform by DFT.

S1.2 Boundary smoothing

The metamaterial unit cells resulting from the above procedure are defined on a pixel grid, which implies non-smooth surfaces and sharp corners. This is undesirable due to the challenges it poses for manufacturability, because of stress concentrations and resulting numerical convergence problems, and due to numerical challenges of the finite element (FE) solver, especially when pixelated boundaries come into contact. As a remedy, a post-processing step is applied prior to simulating the mechanical response, which generates a smoothed version of the pixel boundary by using the midpoints of each straight edge of the material domain boundary as control points for a spline curve, as illustrated in Figure S1.

¹We point out that, alternatively, the noise can be formulated to be Hermitian in the Fourier space, ensuring that the corresponding inverse DFT returns a real signal that preserves the power spectrum characteristics. However, we are not further concerned with the statistical characterization of the GRF and hence omit this approach.

Algorithm 1 Metamaterial unit cell generation

```

1: Set even pixel number  $N$ 
2: while generated structure not valid do
3:   Sample complex  $z_{k_1, k_2} \sim \mathcal{N}(0, 1) + i\mathcal{N}(0, 1)$  on centered  $N \times N$  grid
4:   Compute  $F_{k_1, k_2} = z_{k_1, k_2} P(k_1, k_2)$ ,
       where  $P(k_1, k_2) \propto (k_1^2 + k_2^2)^{-\alpha/2}$  with  $\alpha = 3$ 
5:   Transform to  $N \times N$  pixel grid via inverse DFT as
       
$$f_{x_1, x_2} = \frac{1}{N^2} \sum_{k_1=-N/2}^{N/2-1} \sum_{k_2=-N/2}^{N/2-1} F_{k_1, k_2} e^{i\frac{2\pi}{N}x_1k_1} e^{i\frac{2\pi}{N}x_2k_2}$$

6:   Compute real part standardized over all pixels as1
       
$$\bar{f}_{x_1, x_2} = \frac{\text{Re}(f_{x_1, x_2}) - \mu[\text{Re}(f_{\mathcal{X}})]}{\sigma[\text{Re}(f_{\mathcal{X}})]}$$

7:   Sample threshold  $t \sim \mathcal{U}(0, t_{\max})$  with  $t_{\max} = 3/5$ 
8:   for  $x_1 = 1$  to  $N$  do
9:     for  $x_2 = 1$  to  $N$  do
10:      if  $\bar{f}_{x_1, x_2} > t$  then
11:         $\bar{f}_{x_1, x_2} \leftarrow 1$ 
12:      else
13:         $\bar{f}_{x_1, x_2} \leftarrow 0$ 
14:      end if
15:    end for
16:  end for
17:  if  $\bar{f}$  is four-side connected with at least 10% pixel-coverage then
18:    Mirror  $\bar{f}$  sequentially along vertical and horizontal border
19:    return  $\bar{f}$ 
20:  end if
21: end while

```

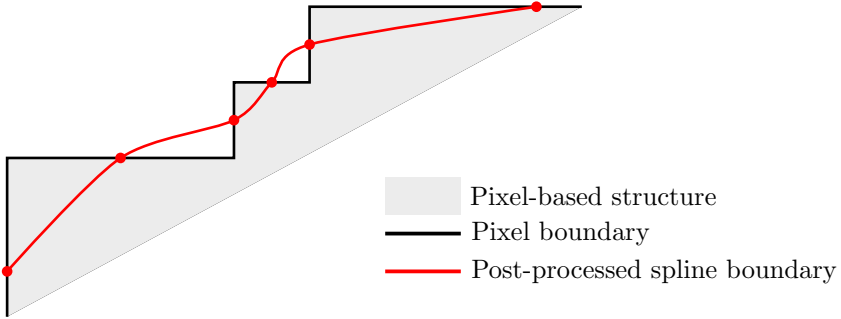


Figure S1: Schematic illustration of the boundary smoothing method, which converts the pixel-based straight boundary segments into a smooth spline defining the surface.

S1.3 Material properties

The base material is described by a visco-elastic-plastic constitutive model, whose elastic moduli and 1D plastic stress-strain response (given by true stress and true plastic strain pairs) are summarized in Table S1. These parameters correspond to the experimentally calibrated model of a thermoplastic acetal homopolymer resin (Delrin) used in 3D printing, as described in [1].

Table S1: Material properties used in finite element simulations.

Property	Value
Elastic Modulus (E)	2.306×10^3 MPa
Poisson's Ratio (ν)	0.35
Damping Coefficient (β)	0.005
Plasticity	MPa, mm/mm
True Stress, True Plastic Strain	40.62, 0.0
	45.24, 0.001133
	52.62, 0.004183
	58.00, 0.0080645
	61.87, 0.012557
	65.81, 0.020035
	69.19, 0.030689
	71.06, 0.038873
	72.61, 0.047114
	73.54, 0.052610
	74.82, 0.06083
	76.74, 0.074477
	78.46, 0.08799
	81.58, 0.11457
	83.00, 0.1276

S1.4 Study of auxetic structures attainable within the chosen design space

The chosen stochastic design sampling based on Gaussian random fields produces general cellular structures and does not introduce any particular geometric features and extreme or peculiar properties commonly explored in metamaterials design. To highlight the generality of that approach, we here include a brief study of potentially auxetic behavior to understand whether the chosen stochastic sampling can generate auxetic unit cells as an example of a mechanical property often found in classical metamaterials[2–4]. We employ the framework provided in [5] to compute the homogenized properties of the pixel-based designs at small strains (setting the material strength under the assumption of linear elasticity to 230 MPa and the void at 0.001 MPa, with a Poisson's ratio of 0.3) and compute the effective Poisson's ratio for the structures in our training dataset (see Section S4). This identified 618 unique structures within the training set with a negative ν_{12} and ν_{21} , five of which

are shown in Fig. S2 below along with their markedly negative Poisson's ratio values.

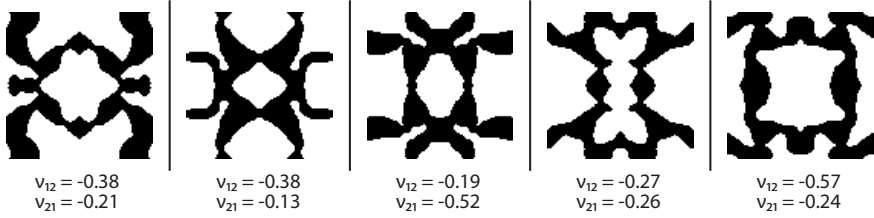


Figure S2: Selection of five auxetic structures from the training dataset with their corresponding Poisson's ratios ν_{12} and ν_{21} (assuming small strains and a linear-elastic base material).

This underscores that our design parametrization, though conceptually simple, can indeed generate intricate structures with properties such as negative Poisson's ratio, which could be leveraged for additional optimization objectives besides the considered stress-strain response.

S2 Theoretical basis of the denoising diffusion process

For completeness, we provide additional details on the derivations presented in the main article. The following results are based on the work of [6, 7], in which further information can be found.

S2.1 Forward diffusion

The *forward* diffusion process is defined by a fixed Markov chain with given variance schedule $\{\beta_t \in (0, 1)\}_{t=1}^T$ as

$$q(\mathbf{x}_{1:T}|\mathbf{x}_0) = \prod_{t=1}^T q(\mathbf{x}_t|\mathbf{x}_{t-1}), \quad q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}), \quad (1)$$

converting data sampled from an unknown data distribution $\mathbf{x}_0 \sim q(\mathbf{x})$ to a standard Gaussian distribution in T steps (i.e., $\beta_T = 1$). Defining $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$, it can be shown[7] that we can sample $\mathbf{x}_t$ at any timestep t in closed form via

$$\begin{aligned} \mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\epsilon}_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\boldsymbol{\epsilon}_{t-2} \\ &= \dots \\ &= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \end{aligned} \quad (2)$$

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where $\boldsymbol{\epsilon}, \boldsymbol{\epsilon}_{t-1}, \dots \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, so that

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}). \quad (3)$$

S2.2 Reverse diffusion

The *reverse* diffusion (also called *denoising*) process $q(\mathbf{x}_{t-1} | \mathbf{x}_t)$, which we approximate by a neural network p parameterized by θ so that $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) \approx q(\mathbf{x}_{t-1} | \mathbf{x}_t)$, allows for sample generation $\mathbf{x}^* \sim q(\mathbf{x})$ by considering the Markov chain in reverse direction as

$$p_\theta(\mathbf{x}_{0:T}) = p(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{x}_t, t), \boldsymbol{\Sigma}(\mathbf{x}_t, t)). \quad (4)$$

To optimize p_θ on a given training dataset assumed to contain representative samples from $q(\mathbf{x})$, the maximization of the (log-)likelihood $\log p_\theta(\mathbf{x}_0) = \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T}$ is computationally infeasible so that we instead maximize the tractable variational lower bound of the log-likelihood (obtained via Jensen's inequality),

$$\log p_\theta(\mathbf{x}_0) \geq \log p_\theta(\mathbf{x}_0) - D_{\text{KL}}(q(\mathbf{x}_{1:T} | \mathbf{x}_0) \| p_\theta(\mathbf{x}_{1:T} | \mathbf{x}_0)), \quad (5)$$

where $D_{\text{KL}}(a \| b)$ denotes the Kullback-Leibler (KL) divergence between two distributions a and b . This can be rewritten as a loss function $\mathcal{L}$ in terms of several KL divergences (and a likelihood term $\mathcal{L}_0$) as

$$\mathcal{L}(\theta) = \mathbb{E} \left[\underbrace{D_{\text{KL}}(q(\mathbf{x}_T | \mathbf{x}_0) \| p_\theta(\mathbf{x}_T))}_{\mathcal{L}_T, \text{ independent of } \theta} + \sum_{t=2}^T \underbrace{D_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))}_{\mathcal{L}_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)}_{\mathcal{L}_0} \right], \quad (6)$$

where $\mathcal{L}_{t-1}$ can be evaluated in closed-form since both q and p_θ are assumed to be Gaussian and since

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}), \quad (7)$$

$$\text{where } \tilde{\boldsymbol{\mu}} = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_t), \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t.$$

Following the work of Ho et al.[7], we further simplify and parameterize the loss not in terms of $\boldsymbol{\mu}$ but the Gaussian noise term $\boldsymbol{\epsilon}$ (which are related via Eq. (7)), so that we obtain the comparably simple loss function

$$\mathcal{L}(\theta)_{\text{simple}} := \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)|]. \quad (8)$$

Note that t is uniformly sampled across the denoising steps and we further condition the model via classifier-free guidance[8] on the corresponding stress-strain response as described in the main article.

S3 Model architecture and computational efficiency

We summarize the most relevant hyperparameters of the diffusion model described in the Methods section in the main article in Table S2.

Table S2: Denoising diffusion architecture hyperparameters.

Hyperparameter	Value
ResNet blocks per down- and upsampling pass	2
ResNet block normalization	Group Normalization[9]
Attention block normalization	Layer Normalization[10]
Activation function	SiLU[11]
Feature map resolutions (downsampling pass)	$96 \times 96 \rightarrow 48 \times 48 \rightarrow 24 \times 24 \rightarrow 12 \times 12$
Latent dimensions (in feature maps)	$64 \rightarrow 128 \rightarrow 256 \rightarrow 512$
Attention head dimension	32
Number of attention heads	8
Positional encoding	Relative & rotary embeddings[12, 13]

To estimate the efficiency of our approach, we provide a detailed overview of the computational runtimes required for the data generation, training and use of our framework in Table S3. Generating a new sample takes the model approximately one minute on a single high-end consumer-grade GPU.

Table S3: Runtime, software, and hardware resources used for different tasks.

¹Reported runtimes are averages only and may vary due to convergence of the FE solver. Dataset generation requires additional simulations since we remove designs that do not converge. ²Tasks were performed on a 2.2 GHz Intel Xeon E5-2650 processor with 256 GB of DDR3 memory at 2.5 GHz. ³Tasks were performed on eight Nvidia Quadro RTX 6000 with 24 GB GDDR6 memory using CUDA 11.7.

Task	Software	Hardware	Runtime
Single design generation and evaluation	Abaqus	CPU (single core) ²	2.5 minutes ¹
Dataset generation	Abaqus	CPU (16 cores) ²	8.1 days ¹
Model training	PyTorch in Python	GPU ³	72 hours
Sample generation	PyTorch in Python	GPU ³	7.2 seconds per sample (averaged over 16 samples)

S4 Training protocol

We provide an overview of the training hyperparameters in Table S4 and the corresponding loss plots in Fig. S3. Since our model has a large number of parameters in comparison to the number of training samples, we expect notable overfitting, which is confirmed in Fig. S3a. We note, however, that this error is an indirect measure of our primary goal, which is to generalize to unseen target stress-strain responses. This is better captured by Fig. S3b, where we condition the model on 16 randomly chosen target responses from the validation data and compare the best relative performance of the corresponding NRMSE (of the FE-reconstructed and target response) with that of the best match. This indicates that the model starts to generalize (i.e., identify designs with better matching responses than the ones existing in the training data) during later stages of the training starting at around 100k iterations, justifying the chosen number of total training iterations.

Table S4: Training hyperparameters.

Hyperparameter	Value
Training data size	53,007
Validation data size	4,924
Batch size	32
Learning rate	10^{-4}
Optimization algorithm	Adam[14]
Iterations	200,000
Exponential Moving Average (EMA) decay	0.995
EMA start (iteration)	2,000
EMA update period (iteration)	10
Diffusion timesteps	256
Condition dropout (for classifier-free guidance[8])	0.1
Dropout	none

S5 Sampling protocol

S5.1 Effective stress response extraction from full-field data

To obtain the effective stress response, we can either sum over the nodal reaction forces (of nodes in contact with the upper or lower rigid surface, which are only defined in the FE simulation but not the forward prediction) or approximate it using the σ_{22} full-field data and exploiting static equilibrium (with some losses in accuracy due to pixel discretization), as illustrated in Fig. S4. For the latter, we evaluate the discrete form of the effective stress at row i as $\sigma_{\text{eff},i} = \int \sigma_{22}(x_1, x_2 = i) dx_1 \approx \frac{1}{j} \sum_{j=1}^{96} \sigma_{22,ij}$, where $\sigma_{22,ij}$ corresponds to the value of the pixel located at row i and column j in the Eulerian frame. We

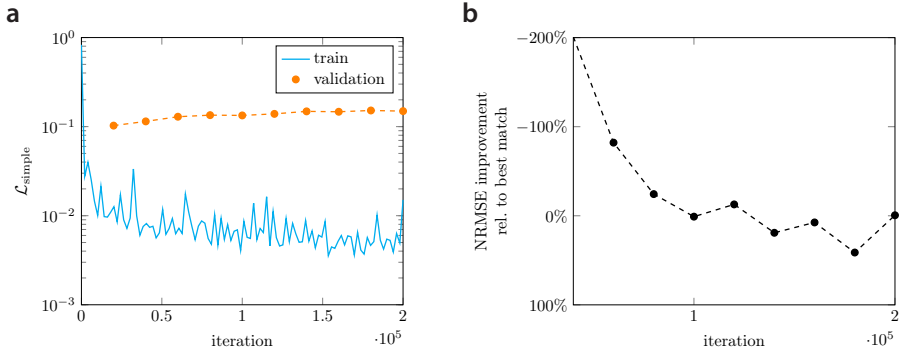


Figure S3: (a) Error convergence evaluated on training and validation data. (b) Relative improvement of the generated design compared to best match in the training data in terms of the NRMSE of the FE-reconstructed and target responses.

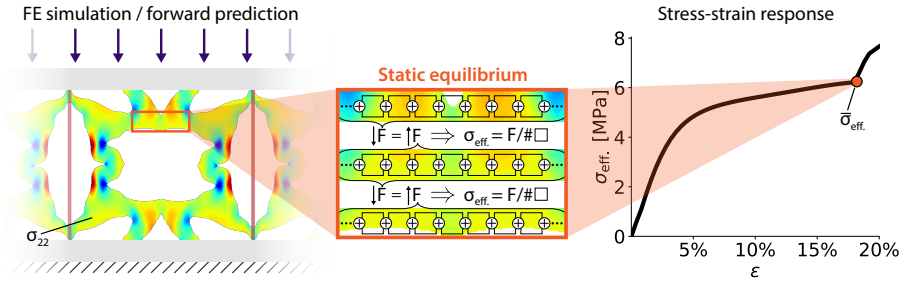


Figure S4: In the considered quasistatic setting, forces must be in equilibrium for any free cut of the structure. Each pixel row can be interpreted as the boundary of such a cut, and we can compute the effective (vertical) stress σ_{eff} of the structure by averaging over all σ_{22} -stress components in a single row of pixels (this sum must match across all rows). We again average over all rows (reducing variance due to our finite pixel grid) to extract a scalar, which is the (effective) stress response $\bar{\sigma}_{\text{eff}}$ of the structure at a certain applied vertical strain. We can equally obtain the effective stress response by collecting the nodal reaction forces at the boundary of our structure, which gives comparable results except for degenerate cases in which the spatial and numerical resolution of the pixels is insufficient, e.g., in case of extremely sparse structures or stress values close to zero.

then extract the effective stress response as $\sigma_{\text{eff.}} = \frac{1}{i} \sum_i \sigma_{\text{eff.},i}$, where i only considers rows in the deformed domain.

The stress distribution in the Lagrangian frame (as predicted by the model) can be linked to the corresponding stress components in the Eulerian frame via a transformation that takes into account the local deformation of each pixel

(precisely, it requires a conversion between Piola-Kirchhoff and Cauchy stress tensor components). During the vertical compression, material points will—on average—extend horizontally, which—in the Eulerian frame—is reflected in an average increase of row-wise pixels with nonzero stresses. As we do not have direct access to the full-field strain data, we here assume that the overall horizontal extension of the configuration is approximately proportional to the overall vertical compression (i.e., the applied vertical strain). This allows us to rescale the predicted effective stress response in the Lagrangian setting in a simplified manner to arrive at the effective (true) stress response in the Eulerian setting as

$$\sigma_{\text{eff.}} \approx \frac{\sigma_{\text{eff., Lagr.}}}{1 - \varepsilon}. \quad (9)$$

This is a simplification since it assumes a simple, uniform proportionality between vertical compression and horizontal extension. Our results, however, indicate that this serves as a good approximation and accurately rescales the predicted response to the actual stress response.

S5.2 Mitigation of accuracy losses due to precision reduction

The inaccuracies arising from the conversion of the full-field data into the `gif` format, as mentioned in Section S4, are addressed as follows. Although we could retain the data in `float` format, the conversion to the `gif` format significantly reduces the storage requirements of the training dataset, which we also envision to be useful for others potentially interested in our (openly released) dataset. The `gif` format stores numerical values in `uint8` format, effectively discretizing each numerical value into one of 256 equidistant bins that span the minimum and maximum values of the respective simulation (across all strain steps). Naturally, this leads to a loss of accuracy in the full-field data, which also holds for the predicted response of the model. While this loss of accuracy is negligible for most generated structures, it can become more pronounced for highly sparse topologies due to the small magnitude of the corresponding stress fields, which will center into very few bins close to zero magnitude.

To mitigate this issue, we proceed as follows. We extrapolate the *predicted* stresses of the model at $\varepsilon = 0\%$ (i.e., no applied strain) by the linear approximation of the small-strain (linear) response, which we obtain via the stresses evaluated at the first two strain steps ($\varepsilon = 0.2\%$ and $\varepsilon = 2\%$). Since stresses must vanish at $\varepsilon = 0\%$, we can shift the effective stress response accordingly, which serves as a simple and effective corrective measure against the introduced inaccuracy. While we apply this procedure to all predicted stress-strain curves, this shift is only noticeable for stress responses of very small magnitude such as in Fig. 3c in the main article.

S6 Further results and full-field data comparison

We here provide the predicted full-field σ_{22} -distribution in the Eulerian frame, including a comparison of the FEM evaluation of the generated structures presented in the main article for all eleven strain steps. This includes four samples of the conditioned generation based on random target curves of the validation data, and also the four samples with more exotic stress responses not closely represented in the training data (see Fig. 3 of the main article). Additionally, we present an initial study of unconditioned sample generation as well as the full-field predictions corresponding to an extremely compliant target response (not closely represented in the training data), and one sample considering buckling, in which the deformation mode between prediction and FE simulation agree.

S6.1 Unconditioned sample generation

We examine the physical accuracy of 100 arbitrarily generated *unconditioned* samples and their deformation mechanisms. The sampled structures should reflect our design space and require the UC to be connected. Interestingly, 96 of the generated structures are indeed connected and adhere to the given symmetries, suggesting that the model has effectively learned these design space features. The NRMSE between the *predicted* and *FE-reconstructed* stress-strain responses is $\epsilon = 5.53\%$ and the relative L_2 -error is $\epsilon_{L_2} = 13.26\%$ (averaged over all samples and strain steps), indicating a strong agreement with FE simulations. The stress distributions for all strain steps of four randomly chosen designs are shown in Fig. S5 and an evaluation of the predicted and actual effective stress response is given in Fig. S6.

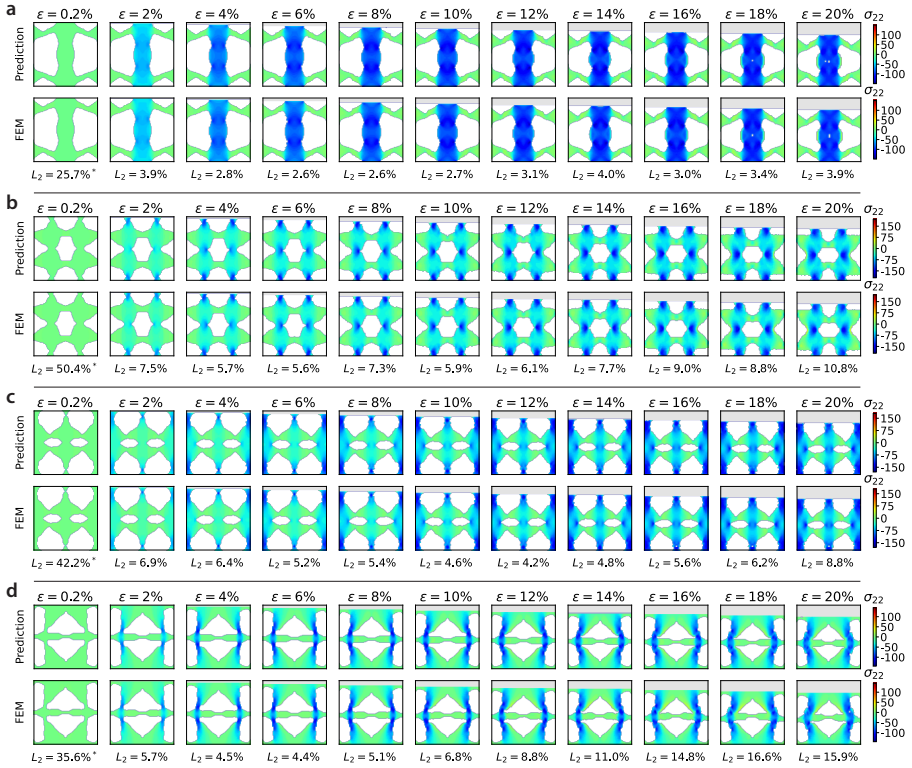


Figure S5: Comparison of the full-field σ_{22} predictions and FE ground truth (indicated in [MPa]) for four generated structures (a-d) without conditioning. Fields are shown in the Eulerian frame for all eleven strain steps, including the corresponding relative L_2 -error. *The relative L_2 -error at $\epsilon = 0.2\%$ is numerically inflated due to the small magnitude of the stress field and is hence not truly indicative (but included for completeness).

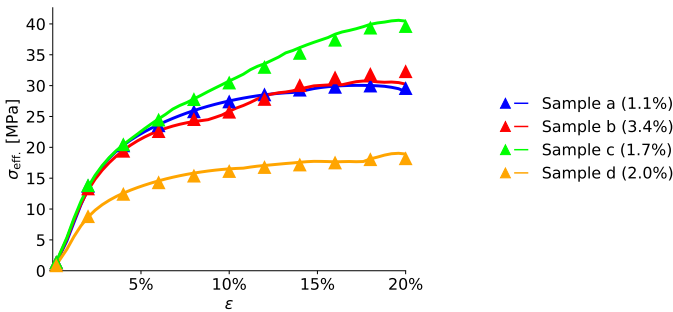


Figure S6: Comparison of the predicted and true effective stress responses σ_{eff} without conditioning. Predicted responses are indicated with $\blacktriangle$ and FE-reconstructed responses with a solid line. The shown responses refer to the four samples presented in Fig. S5, including the corresponding NRMSE of the predicted and FE-reconstructed response (in brackets).

S6.2 Conditioned sample generation

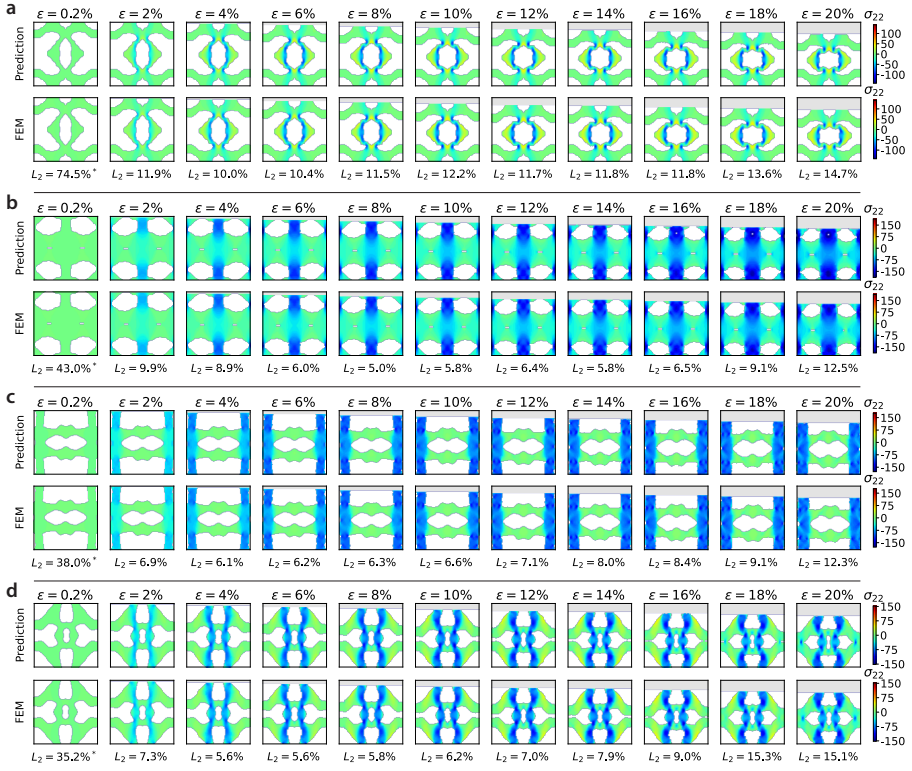


Figure S7: Comparison of the full-field σ_{22} predictions and FE ground truth (indicated in [MPa]) for four generated structures (a-d) with conditioning. Fields are shown in the Eulerian frame for all eleven strain steps, including the corresponding relative L_2 -error. *The relative L_2 -error at $\varepsilon = 0.2\%$ is numerically inflated due to the small magnitude of the stress field and is hence not truly indicative (but included for completeness).

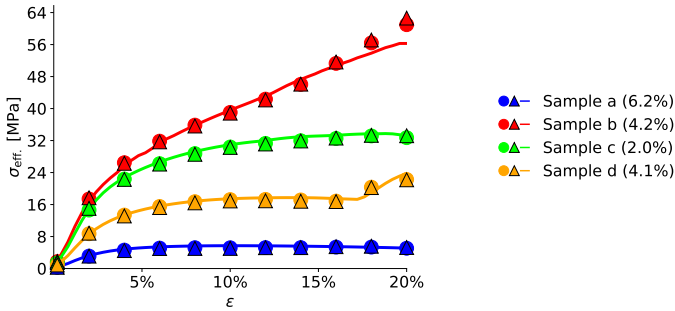


Figure S8: Comparison of the target, predicted, and true effective stress responses $\sigma_{\text{eff.}}$ with conditioning. Target responses are indicated with $\bullet$, predicted responses with $\blacktriangle$, and FE-reconstructed responses with a solid line. The shown responses refer to the four samples presented in Fig. S7, including the corresponding NRMSE of the FE-reconstructed and target response (in brackets).

S6.3 Sample generation for stress-strain responses not represented in the training dataset

Besides the full-field predictions and FE simulation results presented in Fig. S9, we also provide the generated image sequences in video form in Supplementary Videos S1 through S4.

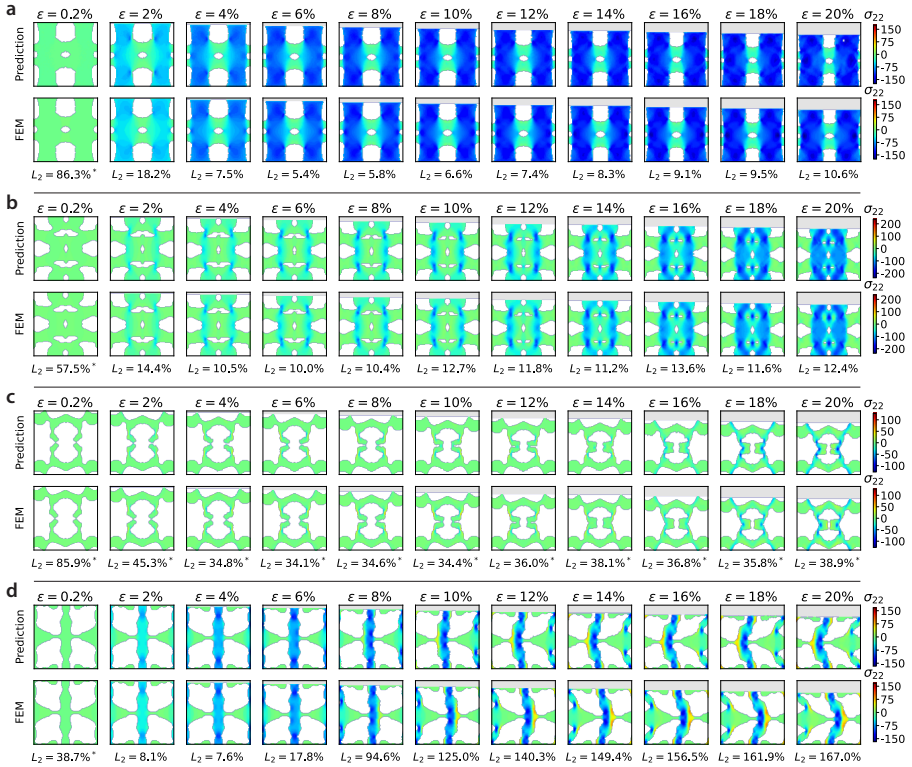


Figure S9: Comparison of the full-field σ_{22} predictions and FE ground truth (indicated in [MPa]) for the four structures presented in Fig. 3 of the main article. Fields are shown in the Eulerian frame for all eleven strain steps, including the corresponding relative L_2 -errors. *The relative L_2 -error is numerically inflated due to the small magnitude of the stress field and is hence not truly indicative (but included for completeness).

S6.4 Effect of dataset and model size

We present a brief study on the influence of the amount of training data and model size. For this, we have explored the performance of the video diffusion model when given a reduced number of training points or with a smaller model parameter count. We trained new models under these constraints using the same training protocol (i.e., with the same number of total iterations) and

evaluated them again as detailed in Section 2.4 of the main article. We present the evaluation in terms of the best NRMSE between FE ground truth and target effective stress σ_{eff} of ten generated samples in Table S5 and Table S6. The results show that for Samples 3b, 3c, and 3d, the original model outperforms the scaled-down versions. Only for Target 3a the other models can identify a design with a performance closer to the match. These variations, given the inherent randomness, are, not fully surprising. Notably, for the challenging Target 3c, only the original model generated a connected design, suggesting it might have captured the underlying design space features more effectively, though it is imaginable that additional samplings from the reduced models might also yield a connected design. Furthermore, a decrease in model parameters appears to have a greater impact on performance than a reduction in training data.

Table S5: Study of the effect of training dataset size on the performance of the video diffusion model. We present the best match of ten generated designs in terms of the NRMSE between the FE evaluation and target responses presented in Fig. 3 (‘Target 3a-d’) of the main article, with the best overall match across all models indicated in **bold**.

Share of full training data	Target 3a	Target 3b	Target 3c	Target 3d
25%	1.15%	3.83%	N/A	2.96%
50%	1.62%	3.77%	N/A	3.07%
100% (original model)	1.49%	1.43%	14.13%	2.40%

Table S6: Study of the effect of overall model parameters on the performance of the video diffusion model. We reduce parameters by reducing the dimension of the initial latent dimension 40 and 32 (see Supplementary Information Table S2). We present the best match of ten generated designs in terms of the NRMSE between the FE evaluation and target responses presented in Fig. 3 (‘Target 3a-d’) of the main article, with the best overall match across all models indicated in **bold**.

Share of full model parameters	Target 3a	Target 3b	Target 3c	Target 3d
30.04%	2.50%	4.56%	N/A	6.87%
43.78%	0.67%	5.25%	N/A	3.34%
100% (original model)	1.49%	1.43%	14.13%	2.40%

S6.5 Conditioning on highly compliant target response

In the main article, we present a generated design with a stress-strain response well above the stiffest sample of the training set. Here, we provide a generated design for a target response significantly below the closest match in the training data (i.e., a significantly softer response). We plot the full-field data of the σ_{22} -distribution for all strain steps in Fig. S10 and an evaluation of the predicted and actual effective stress response in Fig. S11, which again significantly outperforms the indicated closest match in the training dataset, highlighting the model's generalization capabilities. We note that for such highly sparse designs with low stress values, the forward prediction of the effective stress loses accuracy due to the fact that the spatial (pixel) resolution is insufficient to accurately represent such structures and due to our discretization into gif format, which leads to reduced accuracy for stress values close to zero (see also Section S5). This is, however, not a fundamental limitation of the model but rather an expected consequence of the chosen pixel number and data pre-processing.

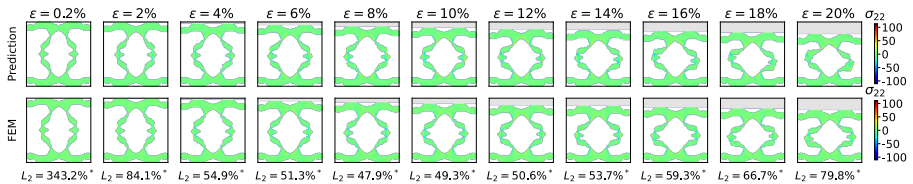


Figure S10: Comparison of the full-field σ_{22} predictions and FE ground truth (indicated in [MPa]) for a sample conditioned with a target response significantly below the most compliant structure in the training set. Fields are shown in the Eulerian frame for all eleven strain steps, including the corresponding relative L_2 -error. *The relative L_2 -error is numerically inflated due to the small magnitude of the stress field and is hence not truly indicative (but included for completeness).

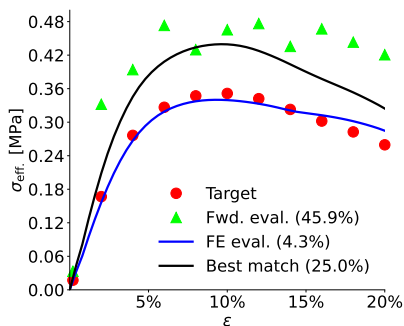


Figure S11: Comparison of target, predicted ('Fwd. eval.'), and FE-reconstructed ('FE eval.') effective stress-strain responses, including the best match in the training dataset ('Best match') for the sample presented in Fig. S10 and including the corresponding NRMSE with respect to the target response (in brackets).

S6.6 Conditioning on buckling

In the main article, we present a generated design for a target response that considers buckling, in which the deformation mode does not agree with the FE simulation. We here provide a generated design for a target response with buckling (at $\varepsilon = 2\%$), for which the deformation mode agrees with the FE simulation. We plot the full-field data of the σ_{22} -distribution for all strain steps in Fig. S12 and an evaluation of the predicted and actual effective stress response in Fig. S13, which again significantly outperforms the indicated closest match in the training dataset, highlighting the model's generalization capabilities.

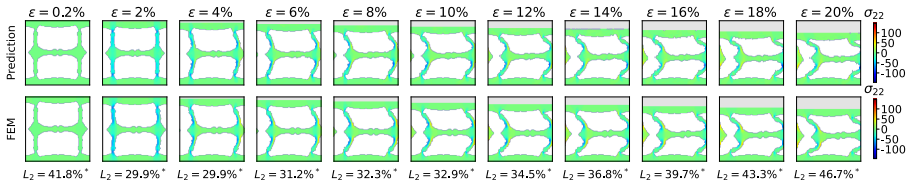


Figure S12: Comparison of the full-field σ_{22} predictions and FE ground truth (indicated in [MPa]) for a sample conditioned with buckling at $\varepsilon = 2\%$. Fields are shown in the Eulerian frame for all eleven strain steps, including the corresponding relative L_2 -error. *The relative L_2 -error is numerically inflated due to the small magnitude of the stress field and is hence not truly indicative (but included for completeness).

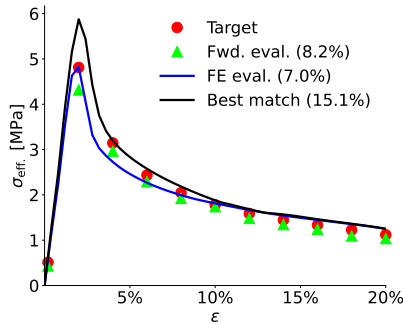


Figure S13: Comparison of target, predicted ('Fwd. eval. '), and FE-reconstructed ('FE eval. ') effective stress-strain responses, including the best match in the training dataset ('Best match') for the sample presented in Fig. S12 and including the corresponding NRMSE with respect to the target response (in brackets).

S6.7 Diversity of generated designs and comparison to the ground truth

To demonstrate the diversity of the generated designs, we present in Fig. S14 and Fig. S15 the ground truth (i.e., the original design) and the top three predictions of our model for the two target responses presented in, respectively, Fig. 3b and Fig. 3d of the main article. This confirms that there is no unique solution to either problem, and the model succeeds in finding a diverse set of designs that all closely match the target response. Apart from the geometrical differences, there are notable variations in the internal stress distributions during deformation, suggesting that the model leverages distinct mechanical mechanisms to achieve the desired output. Furthermore, it is noteworthy that (despite the diverse geometries) all samples consistently outperform—often by a wide margin—the best match found in the training dataset (consisting of 53,007 pairs of designs with corresponding stress-strain responses, as indicated in Table S4).

We emphasize that the non-uniqueness of the sought mapping is a strength of probabilistic model classes, which can handle these inherent ambiguities of the problem space. Probabilistic models are not restricted to (deterministically) predicting a single output (corresponding to a unique solution and thus failing for this setup) but instead provide a range of outputs “sorted” by their associated probabilities. This gives designers the freedom to sample repeatedly and select the design that—in addition to matching the target response—best fulfills secondary objectives, such as manufacturability.

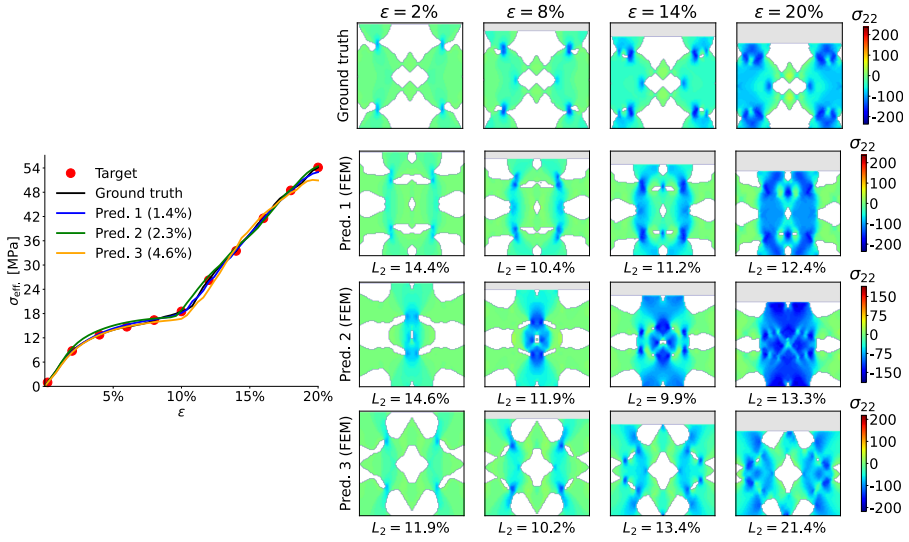


Figure S14: A variety of designs proposed by the diffusion model for a given non-smooth target response. We plot the effective stress response σ_{eff} of the original design of the target response (‘Ground truth’) presented in Fig. 3b of the main article with three synthesized designs from the video diffusion model (‘Pred. 1-3’; NRMSE with respect to the target response in brackets). We also provide the FE evaluation of the original and generated designs in terms of the full-field σ_{22} -distribution (indicated in MPa in the Eulerian frame) and the corresponding relative L_2 -errors of the predicted σ_{22} -distribution, where applicable.

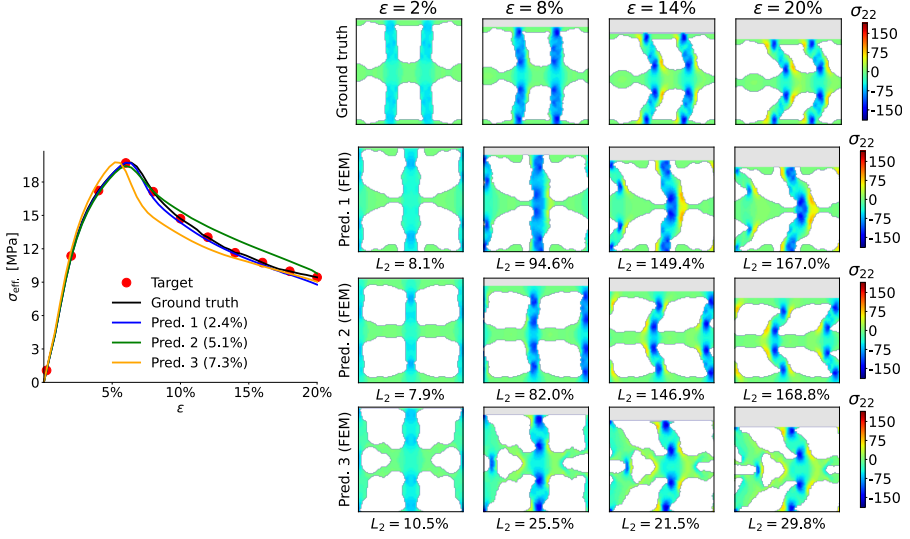


Figure S15: A variety of designs proposed by the diffusion model for a given target response with negative stiffness (buckling). We plot the effective stress response σ_{eff} of the original design of the target response (‘Ground truth’) presented in Fig. 3d of the main article with three synthesized designs from the diffusion model (‘Pred. 1-3’; NRMSE with respect to the target response in brackets). We also provide the FE evaluation of the original and generated designs in terms of the full-field σ_{22} -distribution (indicated in MPa in the Eulerian frame) and the corresponding relative L_2 -errors of the predicted σ_{22} -distribution, where applicable. Note that for Pred. 1 and Pred. 2 the predicted deformation buckles in the opposite direction of the FE evaluation (leading to seemingly high errors).

S6.8 Generalization capabilities of the model

To investigate whether the model generates designs beyond simple “interpolation” of the seen structures (such as simple alterations in relative density), we compare the predicted design to all *optimally scaled* samples in the training data. More specifically, we consider the target response presented in Fig. 3c of the main article and determine the scalar k^* that minimizes the NRMSE, i.e.,

$$k^*(\sigma_{\text{eff.}}^{\text{pred.}}, \sigma_{\text{eff.}}^{\text{true}}) = \arg \min_{k \in \mathbb{R}} \sqrt{\frac{\|k\sigma_{\text{eff.}}^{\text{pred.}} - \sigma_{\text{eff.}}^{\text{true}}\|^2}{\|\sigma_{\text{eff.}}^{\text{true}}\|^2}}, \quad (10)$$

which can be evaluated as

$$k^* = \frac{\sigma_{\text{eff.}}^{\text{pred.}} \cdot \sigma_{\text{eff.}}^{\text{true}}}{\|\sigma_{\text{eff.}}^{\text{pred.}}\|^2}. \quad (11)$$

By this metric, we identified the samples from the training data that perform best in terms of this scaled NRMSE. Interestingly, only three designs surpass or approximately match the NRMSE of our model’s prediction upon optimal scaling. The comparison is shown in Fig. S16 below.

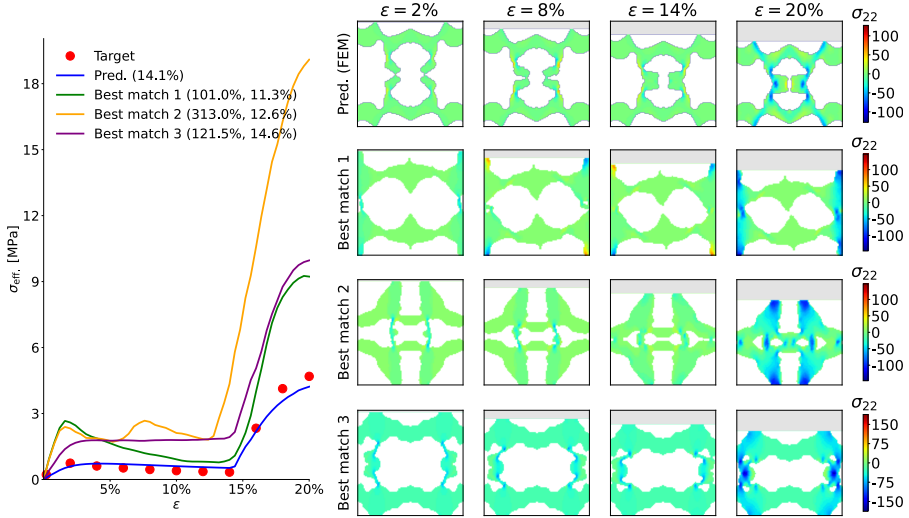


Figure S16: Comparison of best matches in the training data in terms of “*optimally scaled*” responses with the synthesized design. We plot the effective stress response σ_{eff} of the three designs existing in the training data (‘Best match 1-3’; unscaled and scaled NRMSE with respect to the target response in parentheses) with similar (or slightly better) performance compared to the synthesized design (‘Pred.’, unscaled NRMSE with respect to the target response in parentheses) for the target response presented in Fig. 3c of the main article. Best matches are obtained by scaling the response of each sample in the training dataset to minimize the NRMSE with respect to the target response and subsequently selecting the ones with the smallest NRMSE. We also provide the FE evaluation of the full-field σ_{22} -distribution (indicated in MPa in the Eulerian frame) for both the synthesized design and best matches.

It is evident that all “*optimally scaled*” matches differ significantly from the predicted design in both geometry and stress distribution over the applied deformation. From a mechanical standpoint, it is highly improbable that a mere change in relative density would yield similar effective responses. This might be true for some linear responses, but intricate behavior such as buckling or contact necessitates detailed micro-features to match the desired response. Moreover, bending and stretching are two key competing mechanisms in such

compressively loaded structures, which scale quite distinctly with relative density, so that even the linear stiffness properties in low-density cellular structures do not scale linearly with relative density.[15, 16]

Data-driven models typically perform well for interpolation but quickly degrade beyond the support of the training samples, which raises the question if the model is indeed capable of generating “novel” designs and responses, which mainly boils down to the interpretation of this term. Both our results as well as the emerging capabilities of state-of-the-art diffusion models for video generation [17] indicate capabilities far exceeding those that are attainable by straightforward interpolation of the training data (e.g., in terms of mere shape matching and density adjustment). The model encodes intricate mechanical features and their interplay into high-dimensional latent spaces, which allow for non-trivial recombinations of the existing designs, for which no direct assignments to individual data points are possible.

S7 Ablation study

We performed an ablation study to investigate and quantify the effect of training the model on static image data of the initial (linear) response instead of the full image sequence of the entire deformation process. For a meaningful comparison, we consider the identical architecture and dataset for the image-only model but only input the full-field σ_{22} -distribution at the first recorded strain ($\varepsilon = 0.2\%$) along with a binary image of the undeformed structure instead of the $u_{1,2}$ -distribution across the channel dimension during training. The image of the undeformed structure was observed to be necessary to obtain the design from the prediction, since the single full-field distribution here does not allow for reliable extraction. Besides, we must adapt the conditioning since we cannot identify scalar values of the target response with corresponding frames. We therefore select a finer sampling of the target stress-strain response of 51 (instead of 11) values and convert the conditioning via five 1D convolutional layers to a latent representation (similar to the encoding part of a U-Net). We repeat this latent representation to obtain 16 tokens, which we incorporate into the model via the same cross-attention mechanism (and by addition to the diffusion time embedding) as in the video model.

Similar to the study presented in the main article, we assess the model's ability to predict designs aligning with a given target stress-strain response. For this, we once again examine 100 responses of randomly generated designs (unseen during training), sampled with guidance weight $w = 5$. The average NRMSE of the FE-reconstructed response versus the target response is $\epsilon = 19.85\%$, which is considerably larger than that of the model trained on video data ($\epsilon = 6.98\%$). We plot the FE-reconstructed versus target responses of four randomly chosen designs in Fig. S17. Notably, the generated designs tend to match the target response well in the small-strain regime but start to diverge as the strain increases. This underlines the importance of integrating data from the entire deformation trajectory into the model and differentiates the presented approach from existing prior work that trains generative models purely on the corresponding design (which may suffice for linear elastic stiffness).[18–20]

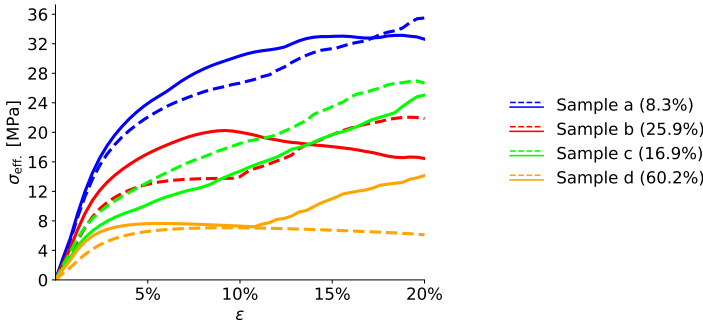


Figure S17: Comparison of target (dashed) and FE-reconstructed (solid line) effective stress-strain responses for four samples conditioned on the response of randomly generated designs generated by an *image* diffusion model, including the corresponding NRMSE of the FE-reconstructed and target response (in brackets). Since the model only generates an image of the predicted small-strain response, we do not obtain a forward prediction over the complete strain path.

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